

SOFTWARE REQUIREMENTS SPECIFICATION

Student Record System

Project: **Student Record System** (Problem Statement No. 12)

Group: Group 6

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1.0	07-09-2026	Group 6	Initial SRS for Student Record System based on the provided SRS references and project statement	

Approvals

Role	Name	Signature / Email	Date
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Table of Contents

1. Introduction
2. Overall Description
3. External Interface Requirement
4. System Features (Detailed)
5. Non-Functional Requirements (Detailed)
6. Quality Attributes & Acceptance Tests
7. System Models and Diagrams
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements, interfaces, system behaviour, and verification criteria for the Student Record System. The system is intended to provide a centralised web-based platform for maintaining student information and academic records, managing courses and enrollment, recording attendance, handling grades and results, tracking fees, generating academic reports, and enforcing controlled access to institutional records.

1.2 Scope

The Student Record System covers role-based authentication for Students, Faculty, and Administrators/Registrar staff; creation and maintenance of student profiles; course catalogue and semester-wise enrollment; attendance recording and monitoring; internal and end-semester marks entry, approval and publication; automatic GPA/CGPA calculation; transcript and mark-sheet generation; fee invoicing and online payment; academic and administrative reporting; user/role administration; record archival; and audit logging.

Out of scope for this release are offline/desktop clients, direct integration with external Learning Management Systems beyond optional data export, and management of hostel or library records.

1.3 Audience

The intended audience includes developers, QA engineers, UI/UX designers, students, faculty members, academic administrators/registrar staff, project guides, and evaluators.

1.4 Definitions and Acronyms

Term	Meaning
SRS	Software Requirements Specification
SRN	Student Registration Number
UI	User Interface
API	Application Programming Interface
JWT	JSON Web Token
OTP	One-Time Password
TLS	Transport Layer Security
RBAC	Role-Based Access Control
GPA	Grade Point Average
CGPA	Cumulative Grade Point Average
RPO	Recovery Point Objective
RTO	Recovery Time Objective
WCAG	Web Content Accessibility Guidelines
RTM	Requirements Traceability Matrix

2. Overall Description

2.1 Product Perspective

The Student Record System is a standalone, cloud-hosted web application accessible through desktop and mobile browsers. It maintains structured academic and administrative records in a managed relational database and stores uploaded documents in object storage. The system can integrate with a Payment Gateway for online fee collection and a Notification Service for email/SMS alerts such as fee reminders, attendance shortfall notifications, and result-publication notices.

2.2 Major Product Functions

- Register, authenticate, and manage role-based user accounts.
- Create, view, and update student profiles and academic records.
- Define courses and manage semester-wise student enrollment.
- Record attendance and calculate course-wise attendance percentages.
- Enter, review, approve, and publish internal and end-semester grades.
- Automatically calculate semester GPA and cumulative CGPA.
- Generate and download official transcripts and mark sheets.
- Generate fee invoices, accept online payments, and track outstanding dues.
- Produce academic, attendance, enrollment, and fee reports.
- Manage users, roles, permissions, record lifecycle, and audit logs.

2.3 User Roles and Characteristics

Role	Responsibilities / Expectations
Student	Views profile, enrolled courses, attendance, published grades, transcripts and fee status; updates permitted contact details; pays fees online; raises grade-review requests.
Faculty	Views course rosters; records attendance; enters and submits marks for courses taught; accesses relevant academic reports.
Administrator / Registrar	Maintains student and course master data; manages roles; verifies profile changes; approves/publishes results; manages fee structures; generates reports; archives records; reviews audit logs.
External Payment Gateway	Processes online fee transactions and sends payment-confirmation callbacks.
Notification Service	Delivers account credentials, password-reset OTPs, fee reminders, attendance alerts and result-publication notifications.

2.4 Operating Environment

Client devices include desktops, laptops, tablets and smartphones running modern versions of Chrome, Firefox, Safari or Edge. The backend is intended to be cloud-hosted, containerised and scalable behind a load balancer, with a managed relational database and object storage.

2.5 Constraints

- Institutional data-protection policies and applicable privacy requirements must be followed.
- The interface should conform to WCAG 2.1 AA accessibility requirements.

- Approved cryptographic libraries must be used for security-sensitive operations.
- Fee payment depends on availability and behaviour of the third-party Payment Gateway.
- Notifications depend on the external Notification Service.
- Enrollment and result processing are governed by institutional academic-calendar windows.

3. External Interface Requirements

3.1 User Interfaces

The primary interface shall be a responsive web application with role-specific dashboards. The Student portal shall provide profile, course, attendance, grades, transcript and fee views. The Faculty console shall provide roster, attendance and grade-entry screens. The Administrator console shall provide master-data management, approvals, reports, user/role management and audit-log views.

- Screen-reader-friendly labels and semantic controls.
- Keyboard navigation for major workflows.
- Clear validation messages for invalid or incomplete data.
- High-contrast and accessible visual presentation.
- Responsive layouts for desktop, tablet and smartphone screens.

3.2 Hardware Interfaces

- Standard desktop, laptop, tablet or smartphone with a supported web browser.
- No specialised hardware is required for normal system operation.

3.3 Software Interfaces

Interface	Purpose	Communication
Payment Gateway API	Online fee payments and payment-confirmation callbacks	HTTPS/TLS
Notification Service API	Email/SMS alerts, credentials and OTP delivery	HTTPS/TLS
Object Storage API	Student photographs, ID proofs and scanned certificates	HTTPS/API
Report Generation	PDF transcripts/mark sheets and CSV/Excel administrative exports	Application service

3.4 Communications

- TLS 1.2 or higher shall be enforced for client-server and service-to-service communication.
- External service calls should use retry/backoff for transient failures.
- Large reports, bulk notifications and other long-running jobs should use asynchronous/queued processing.

4. System Features (Detailed)

The following requirements use the identifier pattern STU-F-###. Each requirement has a priority, stakeholder/source, acceptance criterion, test-case reference and dependency information.

4.1 User Registration & Role-Based Authentication

Description: Allow the institution to provision accounts and support secure role-based login for Student, Faculty and Administrator roles.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-001	The system shall allow an administrator to create accounts for students and faculty, assign a role (Student/Faculty/Administrator), and dispatch auto-generated credentials via the Notification Service.	Type: Functional Priority: High Source / Stakeholder: Administrator	AC-STU-F-001: New account receives login credentials by email within 2 minutes. Test: TC-Auth-01	Notification Service
STU-F-002	The system shall authenticate users using registration/employee ID and password, and issue a signed session token (JWT) on successful login.	Type: Functional Priority: High Source / Stakeholder: Security	AC-STU-F-002: Correct credentials return a valid token; incorrect credentials are rejected with a generic error. Test: TC-Auth-02	
STU-F-003	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and record an audit event.	Type: Functional Priority: High Source / Stakeholder: Security	AC-STU-F-003: After 5 failures, further attempts are blocked and a lockout message is displayed. Test: TC-Auth-03	Audit log
STU-F-004	The system shall allow a user to reset a forgotten password via their registered email using a time-limited OTP.	Type: Functional Priority: Medium Source / Stakeholder: Student / Faculty	AC-STU-F-004: OTP expires after 10 minutes; a valid OTP allows a new password to be set. Test: TC-Auth-04	Notification Service

4.2 Student Profile Management

Description: Maintain the authoritative student profile, including personal, contact and academic-program details.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-010	The system shall allow an administrator to create and maintain the master student profile, including SRN, personal details, and program/branch/semester.	Type: Functional Priority: High Source / Stakeholder: Administrator	AC-STU-F-010: Saved profile is retrievable by SRN and shows all mandatory fields. Test: TC-Prof-01	
STU-F-011	The system shall allow a student to view their profile and update permitted fields (contact number, address, emergency contact), subject to administrator verification.	Type: Functional Priority: Medium Source / Stakeholder: Student	AC-STU-F-011: Submitted changes appear as 'Pending Verification' until approved by an administrator. Test: TC-Prof-02	STU-F-073
STU-F-012	The system shall allow an administrator to upload and attach scanned documents (photograph, ID proof) to a student profile.	Type: Functional Priority: Medium Source / Stakeholder: Administrator	AC-STU-F-012: Uploaded documents are stored and viewable from the profile within 30 seconds. Test: TC-Prof-03	Object storage
STU-F-013	The system shall maintain a complete change history (audit trail) of edits made to a student profile, including editor and timestamp.	Type: Functional Priority: Medium Source / Stakeholder: Compliance	AC-STU-F-013: Profile history lists all prior changes with editor and time. Test: TC-Prof-04	

4.3 Course & Enrollment Management

Description: Define the course catalogue and manage semester-wise student enrollment within institutional limits.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-020	The system shall allow an administrator to define a course with code, title, credits, semester and department.	Type: Functional Priority: High Source / Stakeholder: Administrator	AC-STU-F-020: Saved course appears in the catalogue and is selectable during enrollment. Test: TC-Enr-01	
STU-F-021	The system shall allow a student to enroll in courses offered for their semester within the configured enrollment window, subject to prerequisite checks.	Type: Functional Priority: High Source / Stakeholder: Student	AC-STU-F-021: Enrollment succeeds only if prerequisites are met and the window is open; otherwise a clear error is shown. Test: TC-Enr-02	
STU-F-022	The system shall allow a student to view their current semester course list and total credit load.	Type: Functional Priority: Medium Source / Stakeholder: Student	AC-STU-F-022: Course list and credit total match enrollment records. Test: TC-Enr-03	
STU-F-023	The system shall prevent enrollment beyond the maximum permissible credit limit configured per semester.	Type: Functional Priority: High Source / Stakeholder: Academic Policy	AC-STU-F-023: Attempt exceeding the limit is rejected with an explanatory message. Test: TC-Enr-04	STU-F-020
STU-F-024	The system shall allow faculty to view the roster of students enrolled in a course they teach.	Type: Functional Priority: Medium Source / Stakeholder: Faculty	AC-STU-F-024: Roster reflects enrollment as of the last enrollment-window close. Test: TC-Enr-05	

4.4 Attendance Management

Description: Record session-wise attendance and track cumulative attendance against institutional thresholds.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-030	The system shall allow faculty to record daily/session-wise attendance for each student enrolled in their course.	Type: Functional Priority: High Source / Stakeholder: Faculty	AC-STU-F-030: Attendance is saved and cannot be duplicated for the same date/session. Test: TC-Att-01	
STU-F-031	The system shall calculate and display cumulative attendance percentage per course per student, updated within 5 seconds of entry.	Type: Functional Priority: High Source / Stakeholder: Student / Faculty	AC-STU-F-031: Percentage recalculates correctly after each new entry. Test: TC-Att-02	STU-F-030
STU-F-032	The system shall flag and notify students whose attendance falls below the configured minimum threshold (e.g. 75%).	Type: Functional Priority: Medium Source / Stakeholder: Academic Policy	AC-STU-F-032: Notification is sent within 1 hour of crossing below the threshold. Test: TC-Att-03	Notification Service
STU-F-033	The system shall allow a student to view their attendance record for each enrolled course.	Type: Functional Priority: Medium Source / Stakeholder: Student	AC-STU-F-033: Displayed attendance matches faculty-submitted records. Test: TC-Att-04	

4.5 Grades, Marks & Transcript Management

Description: Support entry, approval and publication of grades, automatic GPA/CGPA computation, and transcript generation.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-040	The system shall allow faculty to enter and submit internal assessment and end-semester marks for students in courses they teach.	Type: Functional Priority: High Source / Stakeholder: Faculty	AC-STU-F-040: Submitted marks are saved against the correct course/student and marked 'Pending Approval'. Test: TC-Grd-01	
STU-F-041	The system shall allow an administrator to review and approve/publish results before they become visible to students.	Type: Functional Priority: High Source / Stakeholder: Administrator	AC-STU-F-041: Results are visible to students only after administrator approval. Test: TC-Grd-02	STU-F-040
STU-F-042	The system shall automatically compute semester GPA and cumulative CGPA using the configured credit-weighted grading scheme upon result publication.	Type: Functional Priority: High Source / Stakeholder: Academic Policy	AC-STU-F-042: GPA/CGPA matches manual calculation for a sample of 10 test records. Test: TC-Grd-03	STU-F-041
STU-F-043	The system shall allow a student to view published grades and download an official transcript/mark sheet as a PDF.	Type: Functional Priority: High Source / Stakeholder: Student	AC-STU-F-043: PDF matches published grade data and includes institutional letterhead. Test: TC-Grd-04	
STU-F-044	The system shall allow a student to raise a grade-review/revaluation request within a configured window after result publication.	Type: Functional Priority: Medium Source / Stakeholder: Student	AC-STU-F-044: Requests after the window closes are rejected automatically. Test: TC-Grd-05	
STU-F-045	The system shall log every grade entry or modification with faculty ID, timestamp and previous value for audit purposes.	Type: Functional Priority: High Source / Stakeholder: Compliance	AC-STU-F-045: Audit log shows complete ordered history for a sample grade record. Test: TC-Grd-06	

4.6 Fee Management

Description: Generate fee invoices, accept online payments, and manage dues and registration holds.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-050	The system shall generate a semester fee invoice for each enrolled student based on the configured program/course fee structure.	Type: Functional Priority: High Source / Stakeholder: Administrator	AC-STU-F-050: Invoices generated for all enrolled students within 24 hours of enrollment-window close. Test: TC-Fee-01	
STU-F-051	The system shall allow a student to pay fees online via the integrated Payment Gateway and receive an e-receipt.	Type: Functional Priority: High Source / Stakeholder: Student	AC-STU-F-051: Successful payment produces a downloadable e-receipt within 1 minute. Test: TC-Fee-02	Payment Gateway
STU-F-052	The system shall update a student's fee-due status in real time upon a successful payment-confirmation callback from the Payment Gateway.	Type: Functional Priority: High Source / Stakeholder: Payment Gateway	AC-STU-F-052: Fee-due status changes to 'Paid' within 30 seconds of a confirmed callback. Test: TC-Fee-03	STU-F-051
STU-F-053	The system shall send a reminder notification for pending dues a configurable number of days before the due date.	Type: Functional Priority: Medium Source / Stakeholder: Administrator	AC-STU-F-053: Reminder is dispatched on the configured schedule for all students with pending dues. Test: TC-Fee-04	Notification Service
STU-F-054	The system shall restrict next-semester registration if outstanding dues exceed the configured threshold, unless overridden by an administrator.	Type: Functional Priority: High Source / Stakeholder: Academic Policy	AC-STU-F-054: Student with dues above threshold cannot enroll unless administrator override is recorded. Test: TC-Fee-05	STU-F-021

4.7 Reporting & Analytics

Description: Provide consolidated academic, attendance and financial reporting for administrators and faculty.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-060	The system shall allow an administrator or faculty member to generate class-wise/course-wise performance reports (average marks, pass percentage, attendance summary).	Type: Functional Priority: Medium Source / Stakeholder: Administrator / Faculty	AC-STU-F-060: Report matches underlying grade and attendance data for a sample course. Test: TC-Rep-01	
STU-F-061	The system shall allow an administrator to export student records, marks and fee reports in CSV/Excel format.	Type: Functional Priority: Medium Source / Stakeholder: Administrator	AC-STU-F-061: Exported file contains all filtered records with correct column headers. Test: TC-Rep-02	
STU-F-062	The system shall provide an administrator dashboard summarising enrollment counts, attendance trends and fee-collection status.	Type: Functional Priority: Medium Source / Stakeholder: Administrator	AC-STU-F-062: Dashboard figures reconcile with underlying records as of the last refresh. Test: TC-Rep-03	
STU-F-063	The system shall allow generation of a consolidated academic transcript (all semesters) for a graduating student on request.	Type: Functional Priority: High Source / Stakeholder: Administrator	AC-STU-F-063: Transcript lists every published semester result and the final CGPA. Test: TC-Rep-04	STU-F-041, STU-F-042

4.8 Administration & Access Control

Description: Provide role and permission management, record lifecycle management and audit visibility for administrators.

Req ID	Requirement	Type / Priority / Source	Acceptance Criteria / Test	Dependencies
STU-F-070	The system shall allow an administrator to assign or modify user roles and permissions.	Type: Functional Priority: High Source / Stakeholder: Administrator	AC-STU-F-070: Role change takes effect on the user's next login. Test: TC-Adm-01	
STU-F-071	The system shall enforce RBAC so students cannot access faculty/administrator functions and faculty cannot access another faculty member's course data.	Type: Functional Priority: High Source / Stakeholder: Security	AC-STU-F-071: RBAC test matrix passes for all roles and modules. Test: TC-Adm-02	
STU-F-072	The system shall allow an administrator to deactivate or archive a student record upon graduation, transfer or withdrawal, while retaining historical data for the configured retention period.	Type: Functional Priority: Medium Source / Stakeholder: Administrator	AC-STU-F-072: Archived record is read-only, excluded from active enrollment lists, and remains retrievable. Test: TC-Adm-03	
STU-F-073	The system shall maintain a searchable audit log of all administrative actions (role changes, record deactivation, overrides).	Type: Functional Priority: High Source / Stakeholder: Compliance	AC-STU-F-073: Audit log searchable by actor, action type and date range. Test: TC-Adm-04	

5. Non-Functional Requirements (Detailed)

The following measurable requirements use the identifier pattern STU-NF-###.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
STU-NF-001	95% of page loads (dashboard, profile, course list, transcript) shall complete within 3 seconds under normal load.	Performance	High	Load test shows p95 ≤ 3s. Test: TC-Perf-01
STU-NF-002	The platform shall provide 99.5% availability monthly, excluding scheduled maintenance windows.	Reliability	High	Uptime reports show ≥99.5% per month. Test: Ops reports
STU-NF-003	The system shall support at least 3,000 concurrent users, including peak enrollment and result-publication periods, without degradation beyond defined performance targets.	Scalability	High	Load test at 3,000 concurrent users passes. Test: TC-Perf-02
STU-NF-004	All record modifications (grades, attendance, fee status, profile edits) shall be logged with timestamps and retained for 5 years for audit and accreditation purposes.	Audit / Data Retention	High	Automated archival and retrieval verified. Test: TC-Ops-01
STU-NF-005	The web interface shall conform to WCAG 2.1 AA accessibility guidelines.	Usability / Accessibility	Medium	Accessibility audit pass. Test: TC-UX-01
STU-NF-006	The system shall support data backup with RPO ≤ 24 hours and RTO ≤ 4 hours.	Reliability / DR	Medium	Disaster-recovery drill meets RPO/RTO targets. Test: TC-DR-01

5.1 Security

5.1.1 Security Objectives

- Protect student and staff personal data (PII) and academic records from unauthorized access or disclosure.
- Ensure the integrity of grade and attendance data by preventing unauthorized modification.
- Ensure the confidentiality and integrity of fee-payment transactions processed through the Payment Gateway.
- Maintain a tamper-evident audit trail of all administrative and grade-related actions.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test
STU-SR-001	TLS 1.2+ shall be mandatory for all network connections between client, server, and the Payment Gateway / Notification Service.	Security	High	TLS scan shows no weak ciphers. TC-Sec-01
STU-SR-002	User passwords shall be hashed using a salted, adaptive algorithm (e.g. bcrypt/argon2) and never stored in plaintext.	Security	High	Password-store review confirms hashing. TC-Sec-02
STU-SR-003	The system shall enforce RBAC so that each role can access only its permitted modules and data.	Security	High	RBAC test matrix passes for all roles. TC-Sec-03
STU-SR-004	Published grade and attendance records shall be immutable except via a logged, administrator-approved correction workflow.	Security	High	Direct edits to published records are rejected outside the workflow. TC-Sec-04
STU-SR-005	All administrative and grade-modification actions shall be logged with actor, timestamp and reason, and the log shall be tamper-evident (append-only / checksum-protected).	Security	Medium	Audit-log integrity check passes for a sample of actions. TC-Sec-05

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

Attribute	Expected Quality
Performance	Common pages respond within the specified 3-second p95 target under normal load.
Reliability	The system maintains 99.5% monthly availability and supports backup/recovery objectives.
Scalability	The platform supports at least 3,000 concurrent users during peak academic operations.
Security	RBAC, secure password storage, encrypted communication, controlled modification workflows and audit logging protect institutional records.
Usability	Role-specific dashboards and clear validation messages make common academic workflows straightforward.
Accessibility	The interface conforms to WCAG 2.1 AA requirements.
Maintainability	Modules are separated by responsibility so authentication, profiles, attendance, grades, fees and reporting can be tested and maintained independently.
Auditability	Important changes to grades, attendance, profiles, fees and administrative permissions are traceable to an actor and timestamp.

6.2 Acceptance Criteria

- All High-priority functional requirements are implemented and verified.
- No critical security or reliability requirement has an unresolved failure.
- Mapped test cases in the RTM have been executed and passed for the release.
- Core student workflows—login, profile viewing, enrollment, attendance viewing, grade viewing and fee payment—work end-to-end.
- Core faculty workflows—course roster, attendance entry and marks submission—work end-to-end.
- Core administrator workflows—student/course management, result approval, fee/report administration, role management and audit review—work end-to-end.

6.3 Acceptance Test Suites

- Authentication
- Profile Management
- Course & Enrollment
- Attendance
- Grades & Transcript
- Fee Management
- Reporting
- Administration & Access Control
- Performance
- Security
- Accessibility
- Backup & Recovery

7. System Models and Diagrams

7.1 UML Use-Case Diagram

The use-case model identifies Student, Faculty and Administrator as the primary human actors, with Payment Gateway and Notification Service as external system actors. The core use cases cover authentication, profile management, enrollment, attendance, grades, transcripts, fees, reporting, notifications and administration/access control.

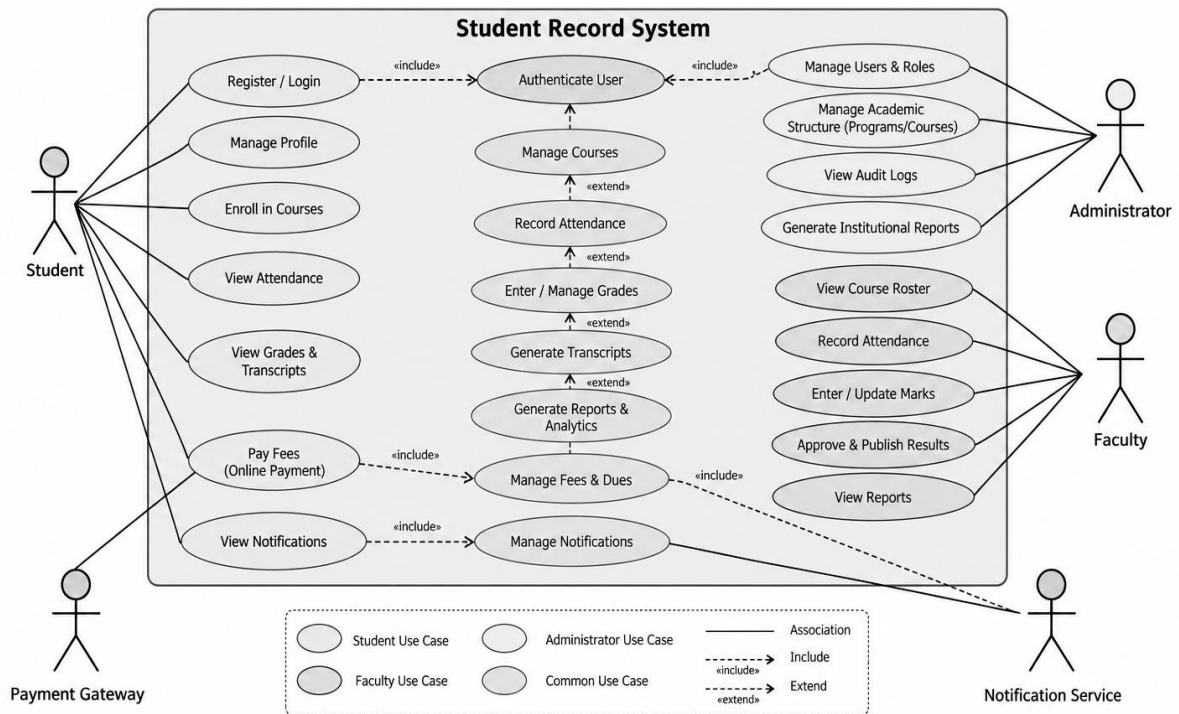


Figure 1. UML Use-Case Diagram for the Student Record System

Figure 1. UML Use-Case Diagram for the Student Record System.

7.2 High-Level System Architecture

The architecture separates the responsive web interface from the application layer and persistence/services. Structured student and academic data is maintained in a relational database, while uploaded documents use object storage. Payment and notification operations are handled through external services.

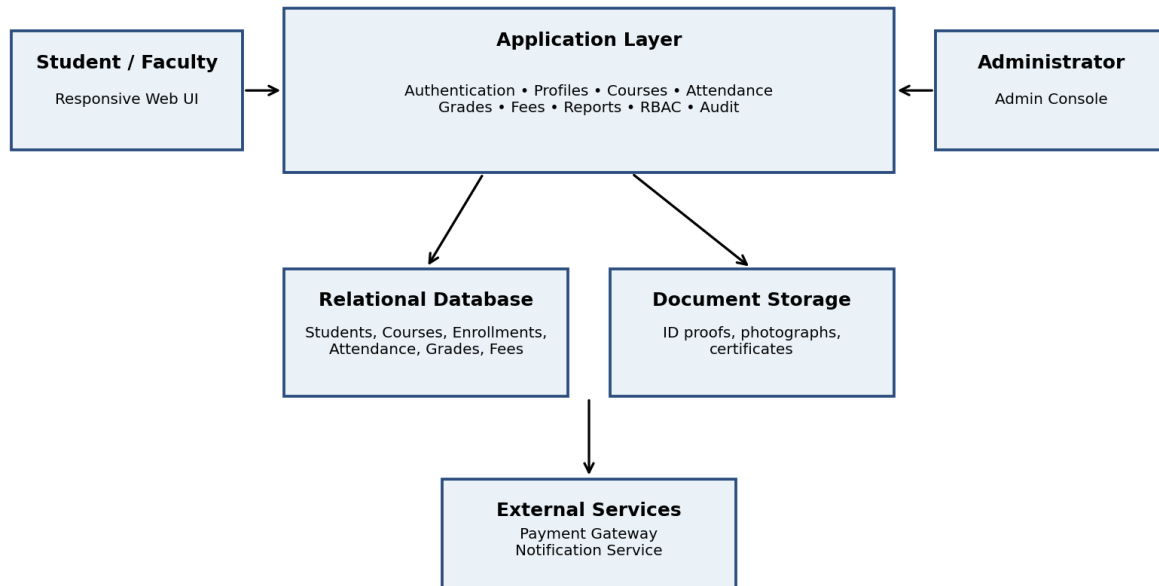


Figure 2. High-Level System Architecture.

7.3 Core Data Entities

Entity	Key Information / Relationship
Student	SRN, personal details, program, branch, semester; linked to enrollments, attendance, grades and fees.
User Account	Login identity, role, credentials/security metadata; linked to Student or Faculty where applicable.
Course	Course code, title, credits, semester, department and prerequisites.
Enrollment	Student-course-semester association and enrollment status.
Attendance	Course/session/student attendance status and derived percentage.
Grade	Student-course marks, grade, status, approval/publish information and history.
Fee Invoice	Student, semester, fee structure, amount due, payment status and receipt.
Document	Uploaded profile document metadata and object-storage reference.
Audit Log	Actor, action, timestamp, previous/current value or reason, and audit metadata.

8. Requirements Traceability Matrix (RTM)

The RTM links representative requirements to their design/module location and test cases. Status N means Not yet executed; P means Passed; A means Accepted after verification.

Req ID	Requirement Short	Section / Design Spec	Module	Test Case(s)	Status	Comments
STU-F-002	User login/authentication	4.1 / DS-Auth-01	AuthModule	TC-Auth-02	N	
STU-F-021	Course enrollment	4.3 / DS-Enroll-01	EnrollmentModule	TC-Enr-02	N	
STU-F-031	Attendance percentage calculation	4.4 / DS-Att-01	AttendanceModule	TC-Att-02	N	
STU-F-041	Result approval & publication	4.5 / DS-Grade-01	GradeModule	TC-Grd-02	N	
STU-F-042	GPA/CGPA computation	4.5 / DS-Grade-02	GradeModule	TC-Grd-03	N	
STU-F-051	Online fee payment	4.6 / DS-Fee-01	FeeModule	TC-Fee-02	N	Payment Gateway
STU-F-073	Administrative audit log	4.8 / DS-Admin-02	AdminModule	TC-Adm-04	N	
STU-NF-001	Page-load performance	5 / DS-Perf-01	WebUI / API Gateway	TC-Perf-01	N	
STU-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-02	N	
STU-SR-004	Immutable published records	5.1.2 / DS-Sec-02	GradeModule	TC-Sec-04	N	

8.1 Requirements Coverage Summary

Requirement Group	IDs Covered	Primary Test Suite
Authentication	STU-F-001 to STU-F-004	Authentication
Student Profiles	STU-F-010 to STU-F-013	Profile Management
Courses & Enrollment	STU-F-020 to STU-F-024	Course & Enrollment
Attendance	STU-F-030 to STU-F-033	Attendance
Grades & Transcripts	STU-F-040 to STU-F-045	Grades & Transcript
Fees	STU-F-050 to STU-F-054	Fee Management
Reporting	STU-F-060 to STU-F-063	Reporting
Administration	STU-F-070 to STU-F-073	Administration
Non-functional	STU-NF-001 to STU-NF-006	Performance / Reliability / UX
Security	STU-SR-001 to STU-SR-005	Security

End of Software Requirements Specification